

SiamRAM: Siamese Tracking with Distractor-Aware Memory and Multi-Phase Reacquisition

System Description

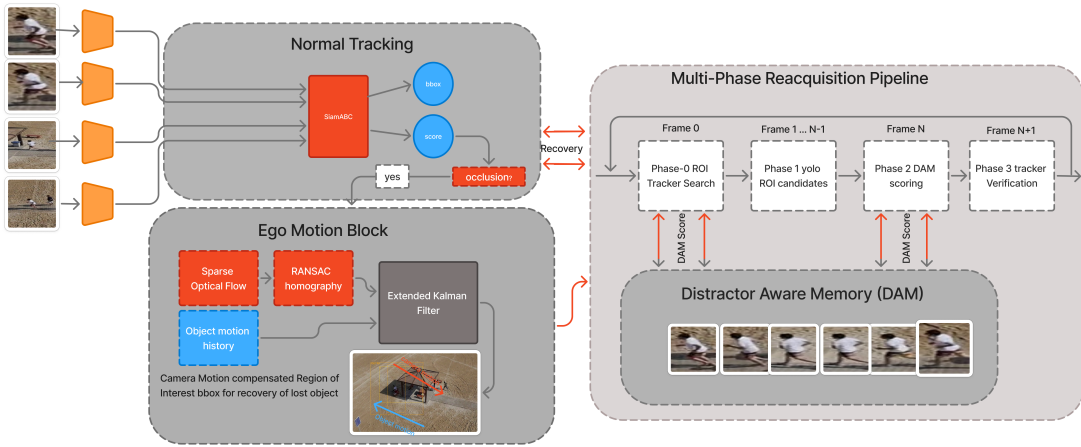


Figure 1: Overall architecture of SiamRAM. Three tightly coupled subsystems: the SiamABC core tracker (top-left), the Ego-Motion Block for camera-motion-compensated Kalman prediction (bottom-left), and the Multi-Phase Reacquisition Pipeline backed by the Distractor-Aware Memory (right).

1 Introduction

Single-object tracking fails in two recurring situations: temporary occlusion and the presence of distractors—objects with a similar appearance to the target. Lightweight trackers drift under occlusion and swap to distractors; heavy memory-augmented systems address both at the cost of real-time latency.

SiamRAM unifies three complementary mechanisms around the SiamABC [3] core tracker: (1) a *Distractor-Aware Memory (DAM)* with a short-term RAM buffer and a stable DRM bank; (2) an *Ego-Motion Block* that compensates for camera motion via sparse optical flow, RANSAC homography, and an EKF [6]; and (3) a *Multi-Phase Reacquisition Pipeline* that cascades through Siamese search, YOLO detection, and DAM-scored verification. A critical prerequisite for all of these mechanisms is that the classification confidence $\hat{c}$ be a *meaningful* signal—genuinely low when the target is absent. Section 3

describes how we restore this property through targeted hard-negative fine-tuning. Section 2 describes the core tracker, Section 4 the memory module, and Section 5 occlusion recovery.

2 Normal Tracking with Siam-ABC

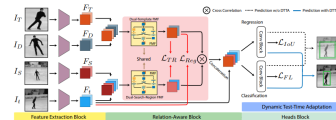


Figure 2: SiamABC normal tracking block. Four crops—static template, dynamic template, search region, and dynamic search—are encoded and fused through attention before the heads produce the classification and regression maps.

2.1 Core Tracker Architecture

SiamABC [3] takes four inputs per forward pass (Fig. 2): a static template z , a dynamic template $\tilde{z}$ updated online, a search region x , and a dynamic search $\tilde{x}$ from the most recent high-confidence frame. All four crops share the same ResNet backbone $\mathcal{E}$ and neck $\mathcal{N}$:

$$f_\bullet = \mathcal{N}(\mathcal{E}(\bullet)), \quad \bullet \in \{z, \tilde{z}, x, \tilde{x}\}. \quad (1)$$

Features are fused through a Fast Parallel Polarized Self-Attention module $\mathcal{A}$:

$$m_z = \mathcal{N}'(\mathcal{A}([f_z; f_{\tilde{z}}])), \quad m_x = \mathcal{N}'(\mathcal{A}([f_{\tilde{x}}; f_x])), \quad (2)$$

and the BoxTower head $\mathcal{T}$ produces a regression map $\hat{\Delta}$ and classification map $\hat{c}$:

$$(\hat{\Delta}, \hat{c}) = \mathcal{T}(m_x, m_z). \quad (3)$$

The final box is decoded at the peak of $\hat{c}$, suppressed by a cosine penalty window at boundaries. As described in Section 3, $\mathcal{T}$ is additionally fine-tuned on hard negatives to sharpen discriminative confidence.

2.2 Dynamic Template Update

A rolling window $\mathcal{M}$ ($M=20$) stores high-quality frames. Frame (I_k, b_k) is admitted when:

$$(s_k \geq \hat{\rho}_k \text{ or } s_k \geq \tau_u) \text{ and } \text{IoU}(b_k, b_{k-1}) \geq \tau_{\text{iou}}, \quad (4)$$

where $\tau_u=0.87$, $\tau_{\text{iou}}=0.6$, and $\hat{\rho}_k$ is a running EMA ($\lambda=0.1$) of past scores. Every $N=10$ frames the best stored frame replaces $\tilde{z}$, provided it exceeds τ_u .

2.3 Adaptive Batch Normalisation at Test Time

Select encoder layers adapt their running statistics to each incoming search crop [5]:

$$\mu_\ell \leftarrow (1-\alpha)\mu_\ell + \alpha\hat{\mu}_\ell, \quad \sigma_\ell^2 \leftarrow (1-\alpha)\sigma_\ell^2 + \alpha\hat{\sigma}_\ell^2, \quad (5)$$

while affine parameters γ_ℓ, β_ℓ remain frozen. This provides implicit adaptation to illumination and sensor-noise shifts at negligible cost.

3 Hard-Negative Fine-Tuning

3.1 The Confidence-Collapse Problem

The pre-trained SiamABC checkpoint exhibits *confidence collapse*: because training used only positive pairs (search crops always containing the target), the head never learned to say “no”. Consequently, $\hat{c}$ saturates near its upper range regardless

of content—background patches, distractors, and occluded regions score as high as the real target. This is catastrophic for SiamRAM: the RAM admission gate (Eq. 7), the occlusion threshold τ_{occ} , and the Phase 0 Siamese check all depend on $\hat{c}$ being a reliable signal.

3.2 Fine-Tuning Protocol

To recover discriminative confidence without disturbing the backbone’s appearance features, we **freeze all ResNet parameters** $\mathcal{E}$ and fine-tune only the BoxTower head $\mathcal{T}$ (and attention necks) for **7 epochs** on UAV footage from UAV123, UAVTrack112, and DTB70. Freezing the backbone prevents catastrophic forgetting and reduces trainable parameters, allowing the head to converge on the hard-negative dataset without over-fitting. Training uses a **50% negative ratio** ($\rho_{\text{neg}}=0.5$): half of all search crops carry an all-zeros classification label, pushing $\hat{c}$ toward zero on those inputs.

3.3 Negative Sampling Strategies

Two complementary strategies (Algorithm 1) construct guaranteed-negative crops. Both apply the same hard geometric gate: a candidate is only accepted if it has *zero pixel overlap* with the target bounding box.

Strategy 1 — Directional Shift (75%). The anchor is pushed away from the target in a random cardinal direction (left, right, up, or down) by at least $(1+\sigma_s+0.1)\times d$ pixels, where $\sigma_s=2.0$ is the search context factor and d is the target dimension on that axis. This clearance guarantees the full context window clears the target boundary. A small lateral jitter ($\pm 0.5d$) is added on the orthogonal axis to prevent the head from learning a directional bias. This strategy produces negatives with the same background texture and lighting as the live sequence.

Strategy 2 — Random Distractor (25%). An anchor of the same size as the target is placed at a uniformly random location in the frame, then checked against the target. This samples background regions that the directional shift never reaches and provides greater textural variety, preventing any residual spatial bias from the repeated cardinal displacements of Strategy 1.

Algorithm 1 Build a hard-negative search crop

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1: for up to 50 attempts do
2:   if Strategy 1 (shift) then
3:     Pick a random direction: left / right / up /
     down
4:     Shift anchor  $\geq (1+\sigma_s+0.1)\times d$  in that direc-
     tion
5:     Add small random jitter on the perpendicular
     axis
6:   else (Strategy 2: random distractor)
7:     Place anchor at a uniformly random  $(x, y)$  in
     the frame
8:   end if
9:   Expand anchor to full context window
10:  if context window does not overlap target then
    ▷ hard gate
11:    return cropped patch ▷ guaranteed negative
12:  end if
13: end for
14: return None                ▷ frame too crowded; skip

```

3.4 Effect on Downstream Components

After 7 epochs, background and distractor crops score well below $\tau_{occ}=0.55$, while genuine target crops retain scores above 0.80. This separation is load-bearing: the RAM admission gate can now exclude distractor-corrupted frames, the occlusion entry criterion fires only on true occlusion, and Phase 0’s $s_t \geq 0.80$ threshold becomes a genuine fast-path. Calibrated confidence is the precondition on which the entire recovery architecture rests.

4 Distractor-Aware Memory (DAM)

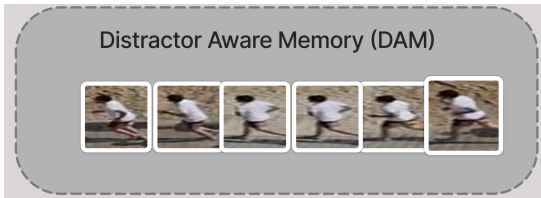


Figure 3: Distractor-Aware Memory. RAM stores geometrically verified short-term hypotheses; stable entries are promoted to the DRM bank for long-occlusion recovery scoring.

The DAM module (Fig. 3) maintains two lightweight descriptor buffers [4]: the RAM (capacity $C_R=50$) holds a short geometric history, and the DRM (capacity $C_D=20$) stores curated stable anchors.

4.1 Appearance Descriptor

Two independently L2-normalised components are computed from a patch P at box $b=(x, y, w, h)$: a 16×16 grayscale texture vector $\hat{\mathbf{p}} \in \mathbb{R}^{256}$ and an $8\times 4\times 4$ HSV histogram $\hat{\mathbf{h}} \in \mathbb{R}^{128}$, combined as:

$$\phi(I_t, b) = \frac{[w_g \hat{\mathbf{p}}; w_c \hat{\mathbf{h}}]}{\|[w_g \hat{\mathbf{p}}; w_c \hat{\mathbf{h}}]\| + \epsilon} \in \mathbb{R}^{384}, \quad w_g=0.4, \quad w_c=0.6. \quad (6)$$

Colour is up-weighted ($w_c > w_g$) for stronger re-identification after background-heavy occlusion.

4.2 RAM Admission and DRM Promotion

RAM admits box b only when geometrically coherent with $\hat{b}_{t-1}$:

$$\text{IoU}(b, \hat{b}_{t-1}) \geq \tau_{\text{ram}} \text{ and } \frac{|a(b) - \bar{a}_t|}{\bar{a}_t + \epsilon} \leq \tau_a, \quad (7)$$

with $\tau_{\text{ram}}=0.40$ and $\tau_a=0.25$. The area gate prevents distractor-driven size jumps from corrupting the buffer.

DRM promotion requires consensus across the last $W=5$ RAM entries:

$$\sum_{j=t-W+1}^t \mathbf{1}[\cos(\phi_j, \phi_t) \geq \tau_{\text{sim}}] \geq m_{\text{min}}, \quad (8)$$

with $\tau_{\text{sim}}=0.60$ and $m_{\text{min}}=3$, filtering entries from brief blur or partial occlusion. Each DRM anchor stores descriptor ψ_k and promotion timestamp ρ_k .

5 Occlusion Handling and Recovery

5.1 Occlusion Entry

The tracker declares occlusion when $s_t < \tau_{occ}=0.55$ persists for $N_p=1$ consecutive frame, at which point the last bounding-box estimate is frozen and the recovery pipeline activates.

5.2 Ego-Motion Compensation

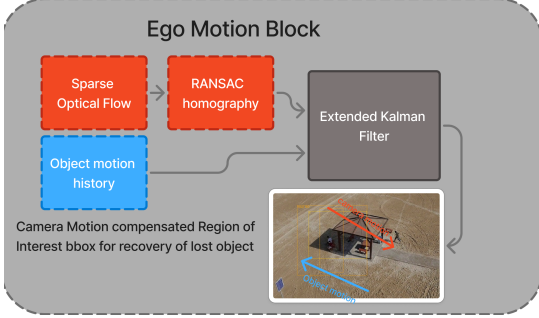


Figure 4: Ego-Motion Block. Sparse LK optical flow feeds RANSAC homography estimation; the resulting transform warps the EKF centre prior to kinematic integration, yielding a camera-motion-robust search window.

Sparse Lucas–Kanade flow [1] on Shi–Tomasi background keypoints yields a 3×3 homography H_t via RANSAC, flagged reliable when the inlier ratio exceeds 0.50. A 4D EKF state $[c_x, c_y, v_x, v_y]^T$ tracks the target centre and velocity. At each step the centre is warped by H_t (when reliable) to remove camera-induced displacement, then advanced by (v_x, v_y) . Box size is smoothed with $\alpha_s=0.85$. During occlusion the EKF runs prediction-only, providing the centre anchor for the search ROI.

5.3 Multi-Phase Reacquisition Pipeline

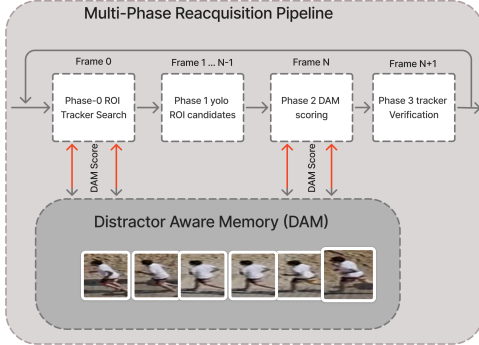


Figure 5: Multi-Phase Reacquisition Pipeline. Phase 0 runs SiamABC in the EKF ROI; Phase 1 collects YOLO candidates; Phase 2 scores via DAM appearance and velocity; Phase 3 verifies with the Siamese tracker.

Once occlusion is declared, SiamRAM runs a four-phase cascade (Fig. 5). Each phase is tried in turn; the first to succeed reinitialises the tracker.

Phase 0 — Siamese ROI Search. SiamABC

is seeded at the EKF-predicted centre within the EKF ROI. If $s_t \geq 0.80$ and a lightweight DRM check passes, recovery succeeds immediately without invoking YOLO.

Phase 1 — YOLO Candidate Collection. YOLOv11n is run on an expanded crop across $F_c=3$ consecutive frames, building per-candidate velocity tracklets. The search margin grows as:

$$\delta_t = \min(\delta_1 \cdot g^{\lfloor n/T_g \rfloor}, \delta_{\max}), \quad (9)$$

with $\delta_1=100$ px, $g=1.4$, $T_g=150$, $\delta_{\max}=500$ px for standard objects, and tighter parameters ($\delta_1=20$, $\delta_{\max}=40$ px) for tiny objects ($< 3\%$ of frame area).

Phase 2 — DAM Appearance and Velocity Scoring. Each candidate c_i is scored against DRM anchors: $\mathcal{S}^*(c_i) = \max_k \cos(\psi_k, \phi(I_t, c_i))$. A velocity consistency term penalises candidates whose motion conflicts with the EKF prediction:

$$\mathcal{V}(c_i) = 0.8 \cdot \frac{\cos(\mathbf{v}_{\text{ekf}}, \mathbf{v}_{c_i}) + 1}{2} + 0.2 \cdot s_{\text{spd}}, \quad (10)$$

where $s_{\text{spd}} = \text{clip}(1 - \|\mathbf{v}_{c_i}\| / \|\mathbf{v}_{\text{ekf}}\| - 1, 0, 1)$, with $s_{\text{spd}}=0$ if $\cos < 0$ and $\mathcal{V}$ clamped to 0.05 when $\cos < -0.5$. The composite score is:

$$\tilde{\mathcal{S}}(c_i) = \mathcal{S}^*(c_i) + 0.20 \cdot \mathcal{V}(c_i). \quad (11)$$

Phase 3 — Tracker Verification. The top candidate is shifted by one EKF-velocity step to reinitialise SiamABC. A score ≥ 0.70 confirms reacquisition; failure returns to Phase 0.

6 Conclusion

SiamRAM integrates online template adaptation, test-time batch normalisation, dual-buffer appearance memory, ego-motion-compensated Kalman filtering, and a four-phase reacquisition cascade. The BoxTower head of the base SiamABC checkpoint exhibited confidence collapse, producing uniformly high scores over backgrounds and distractors alike. We address this by freezing the ResNet backbone and fine-tuning only the head for 7 epochs at a 50% negative ratio. Two geometrically-guaranteed negative strategies—directional shift (75%) and random distractor placement (25%)—together provide complementary background coverage and eliminate spatial biases. Post fine-tuning, $\hat{c}$ cleanly separates target-present from target-absent regions, enabling all downstream confidence-gated logic to function as intended. The velocity consistency term in Phase 2 further discriminates appearance-matched distractors whose kinematics conflict with the EKF prediction, providing a complementary motion-domain signal beyond appearance.

References

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- [2] L. Bertinetto et al., “Fully-convolutional siamese networks for object tracking,” *ECCV Workshops*, 2016.
- [3] *Author(s) TBD*, “SiamABC: Siamese Tracking with Adaptive Batch Calibration,” *arXiv*, 2025.
- [4] S. M. Raza et al., “EdgeDAM: Real-time Object Tracking for Mobile Devices,” *arXiv:2603.05463*, 2026.
- [5] *Author(s) TBD*, “MATA: Motion-Aware Test-Time Adaptation for Visual Tracking,” *arXiv*, 2025.
- [6] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *J. Basic Engineering*, vol. 82, pp. 35–45, 1960.